School of Computer Science Engineering and Information Systems
B.Tech Information Technology
BCSE355L : Cloud Architecture Design
FALL 2026-2027
Course Project -Project Proposal

TEAM NAME : Cloud Busters
Team Member(s) with Reg # and Name : Aarav Gupta ; 24BIT0245 ; 8329103296 ; aarav.rajugupta2024@vitstudent.ac.in Abhishek Yennam ; 24BIT0603 ; 9156284820 ; abhishek.rajesh2024@vitstudent.ac.in Chandru Vel Kumar ; 24BIT0238 ; 6382277008 ; chandru.velkumar2024@vitstudent.ac.in
Project Title: Scalable Multi-Tier Web Application
Abstract : Cloud computing enables organizations to build applications that are scalable, secure, and highly available. This project aims to design and deploy a scalable multi-tier web application using Amazon Web Services (AWS). The application follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Database Layer, ensuring better performance, security, and maintainability. The presentation layer will be hosted on Amazon EC2 instances and distributed using an Elastic Load Balancer (ELB). Auto Scaling Groups will automatically increase or decrease the number of EC2 instances based on incoming traffic to ensure high availability and cost efficiency. The application layer will process user requests, while Amazon RDS will securely manage the relational database. Static resources such as images, CSS, JavaScript files, and documents will be stored in Amazon S3 and delivered efficiently through Amazon CloudFront. To ensure security, the infrastructure will be deployed inside an Amazon Virtual Private Cloud (VPC) using public and private subnets, Security Groups, and IAM roles for controlled access. Amazon CloudWatch will continuously monitor application performance, resource utilization, and system health through logs, metrics, and alarms. The project demonstrates how multiple AWS services can be integrated to build a reliable production style cloud application following AWS best practices.
Objective of the project: • Design and deploy a scalable three-tier web application using AWS cloud services. • Ensure high availability through Elastic Load Balancer and Auto Scaling. • Implement secure networking using Amazon VPC, Security Groups, and IAM. • Store and manage application data using Amazon RDS and Amazon S3.

- Improve content delivery performance using Amazon CloudFront.
- Monitor application health, logs, and resource utilization using Amazon CloudWatch.
- Demonstrate real-world cloud architecture based on AWS best practices.

Innovation component in the project:

Unlike a basic cloud-hosted web application, this project integrates multiple AWS services to create a production-style architecture that automatically scales according to user demand. It combines Auto Scaling, Elastic Load Balancer, CloudFront, CloudWatch, secure VPC networking, and managed database services to provide high availability, security, and performance. The project demonstrates modern cloud deployment practices that are widely used in enterprise Applications.

Technologies:

AWS Services: Amazon EC2, Elastic Load Balancer (ELB), Auto Scaling Group, Amazon RDS, Amazon S3, Amazon CloudFront, Amazon VPC, IAM, Security Groups, Amazon CloudWatch
Development Technologies: HTML, CSS, JavaScript, (Optional: Node.js/Express if implementing a backend), MySQL (through Amazon RDS)

For Evaluation only

Status : **ACCEPTED / REJECTED**

VIT® Vellore Institute of Technology (Deemed to be University under section 3 of UGC Act, 1956)	**School of Computer Science Engineering and Information Systems** **B.Tech Information Technology** **BCSE355L : Cloud Architecture Design** **FALL 2026-2027** **Course Project -Project Proposal** **Review Date: 27.07.2026 – 31.07.2026** **Course Project- Review- I** **Evaluation Sheet**

Title:

Team Name

Project Team

S.No	Register Number	Student Name	Signature	Guided By

Team Member(s) Contribution and Performance Assessment

Components	Student 1	Student 2	Student 3
Understanding Background of & Problem Statement -Abstract (02)			
Objective of the project & Requirements Analysis (02)			
Innovation component in the project (01)			
Total (05)			

Expectation for Next Reviews	Comments

Name & Signature of the Evaluator